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GreenMorph: Sustainable Neuromorphic Computing through Energy-Harvesting and Energy-Driven Online STDP Learning

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Outline

- Introduction
- Proposed Framework
- Evaluation
- Conclusion

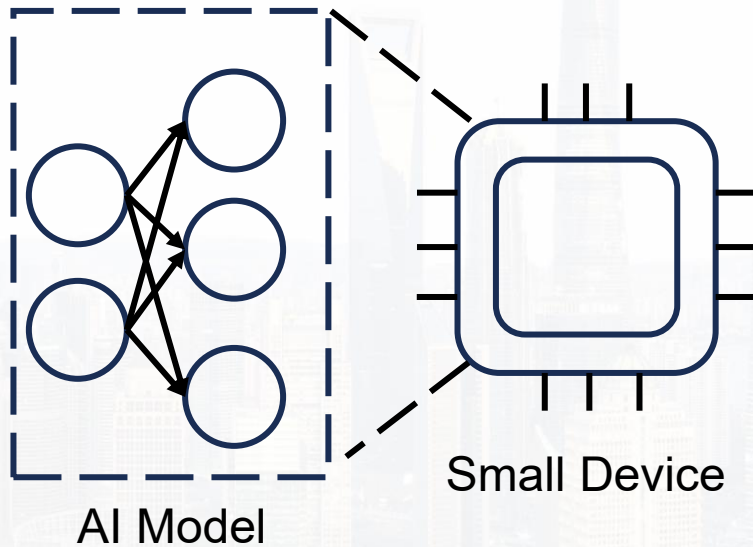
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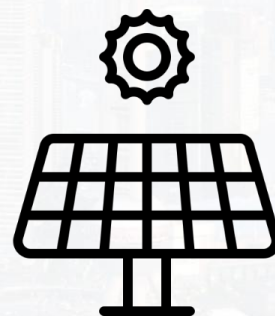
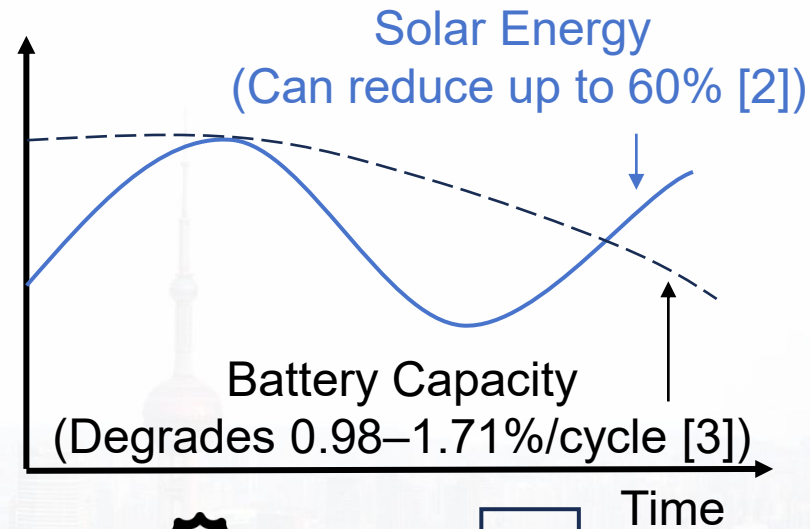
Introduction

Edge AI

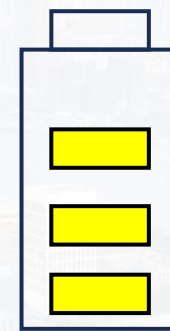
Demand: $\nearrow$ ~+19% annually [1]



Energy Supply



Solar Panel



Battery

Fixed Energy Consumption

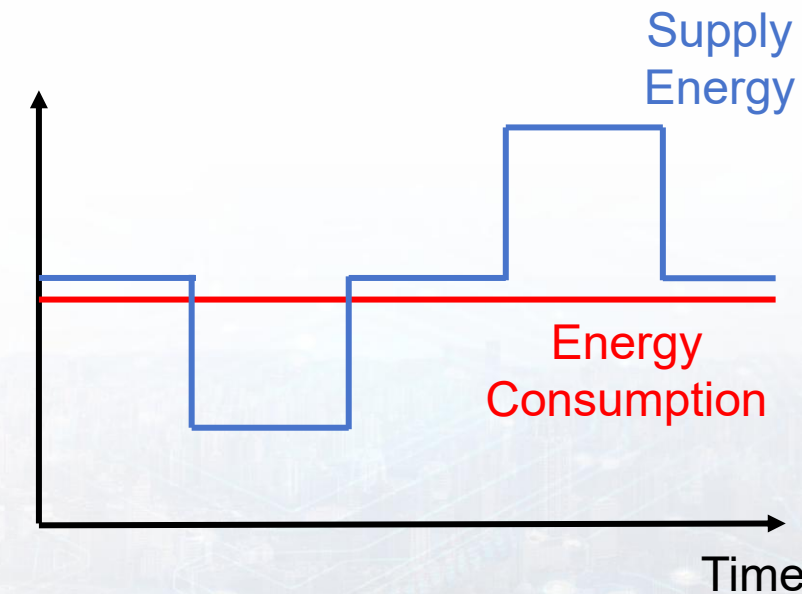
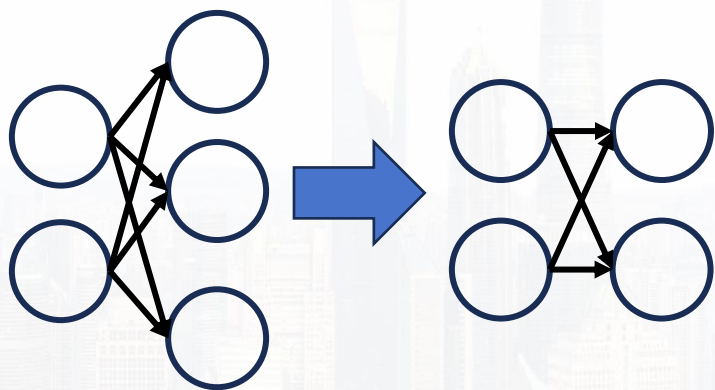


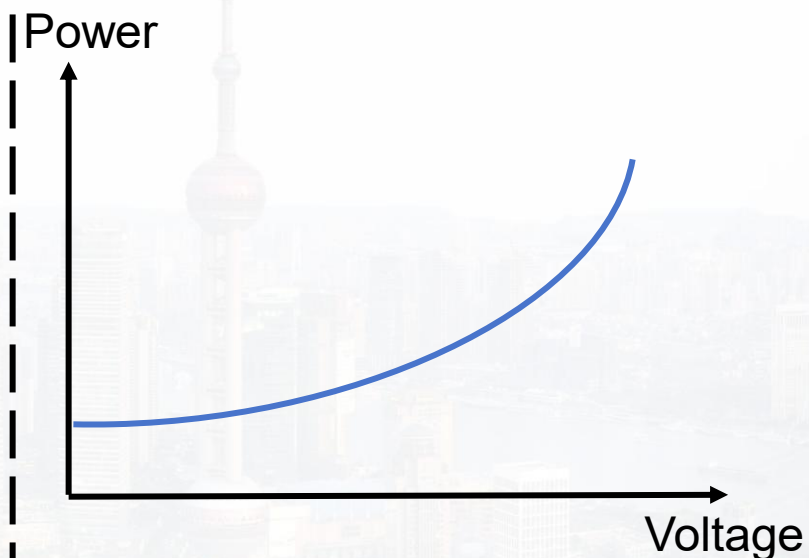
Fig. 1 Edge AI and Energy

Introduction (cnt.)

Model Downscaling [4]



Voltage/Frequency Scaling [5]



Weight Quantization [6]

Floating Point			Integer	
1.2	2.3	→	1	2
1.5	3.1		2	3

Fig. 2 Common Energy Saving Methods

Introduction (cnt.)

Spiking Neural Networks (SNNs) [7]

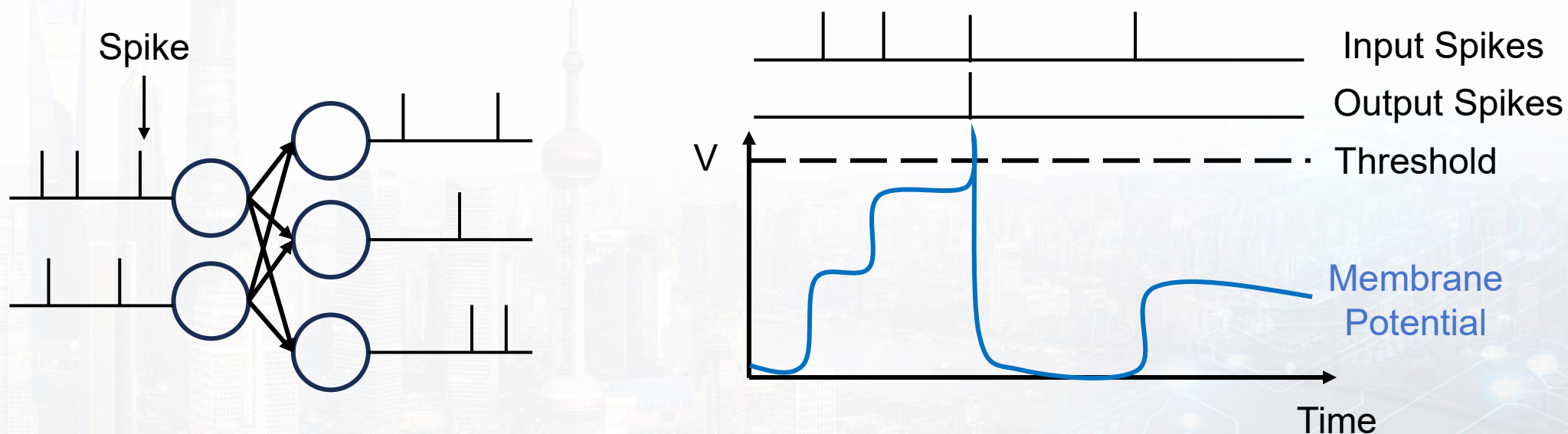


Fig. 3 Spiking Neural Networks

Introduction (cnt.)

Spike-Timing-Dependent-Plasticity (STDP) [8]

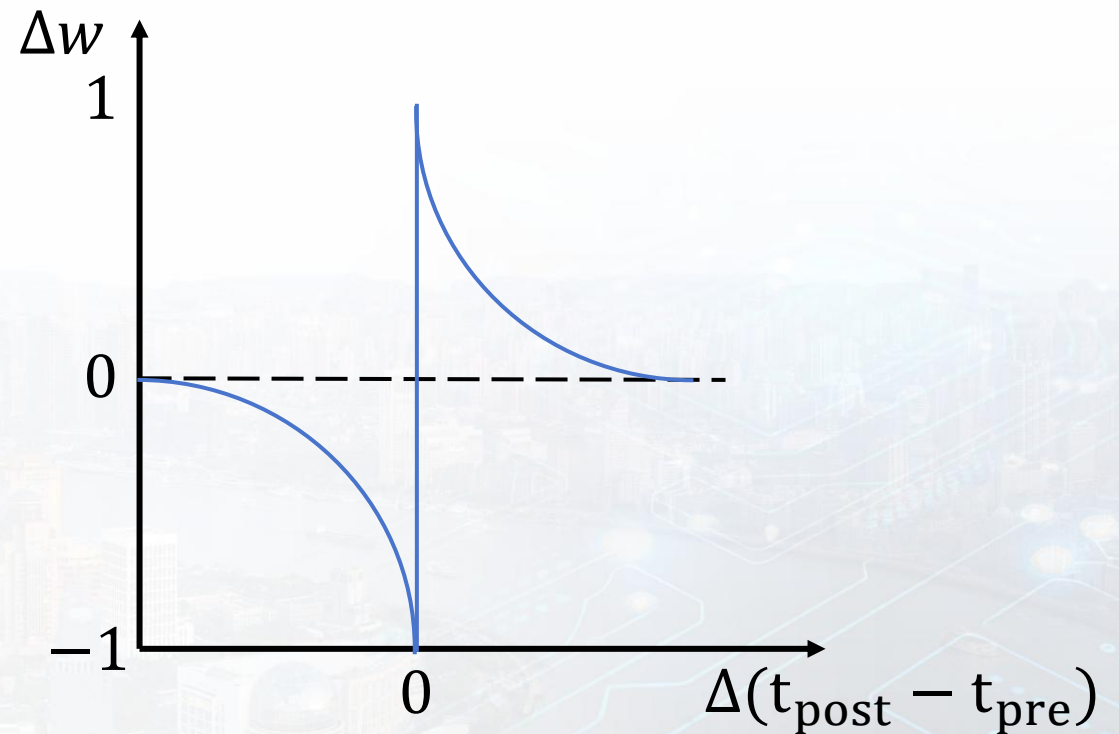
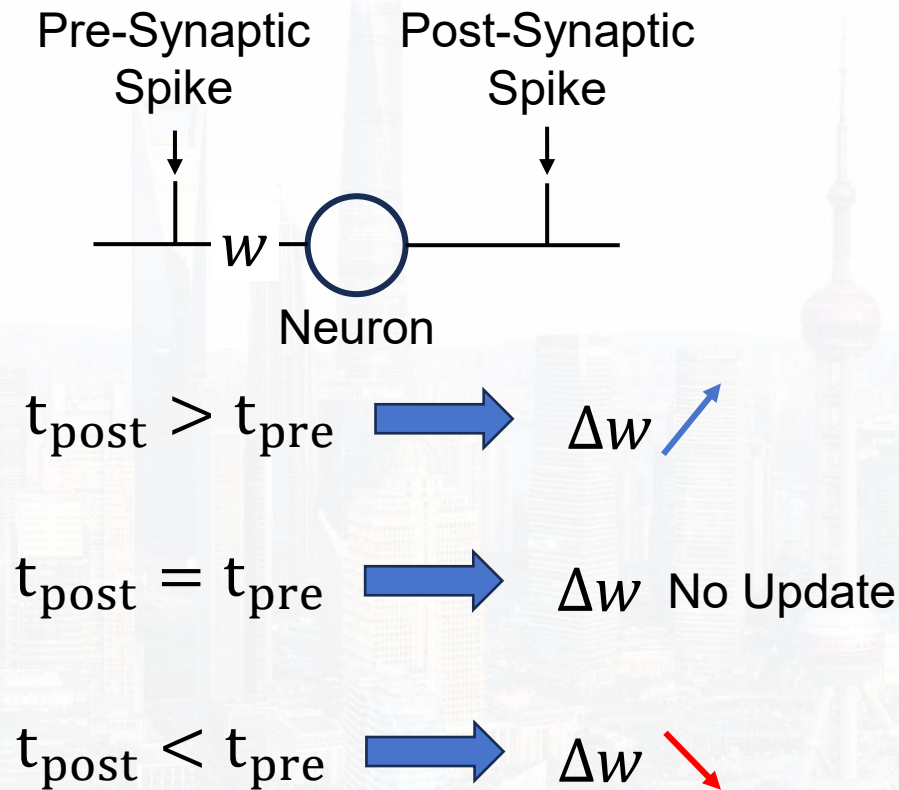


Fig. 4 Spike-Timing-Dependent-Plasticity

Introduction (cnt.)

Challenges:

Unstable energy supply for edge AI systems.

Simple energy reduction leads to suboptimal performance.

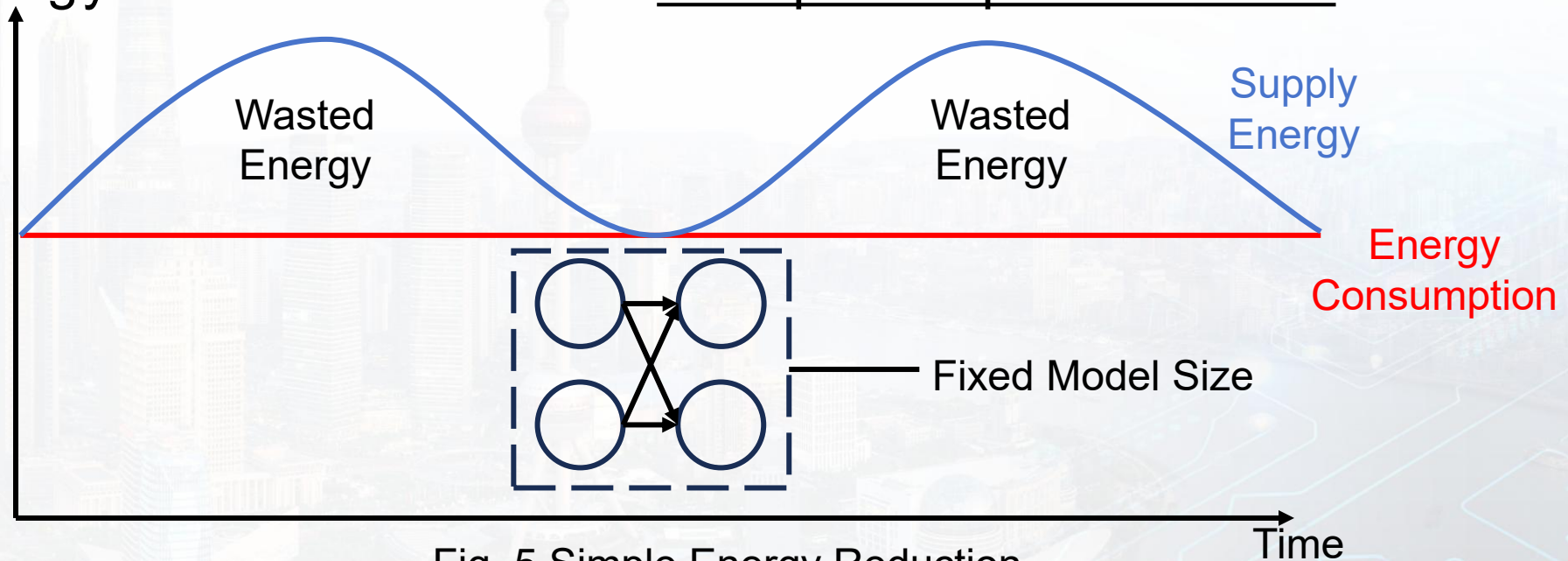


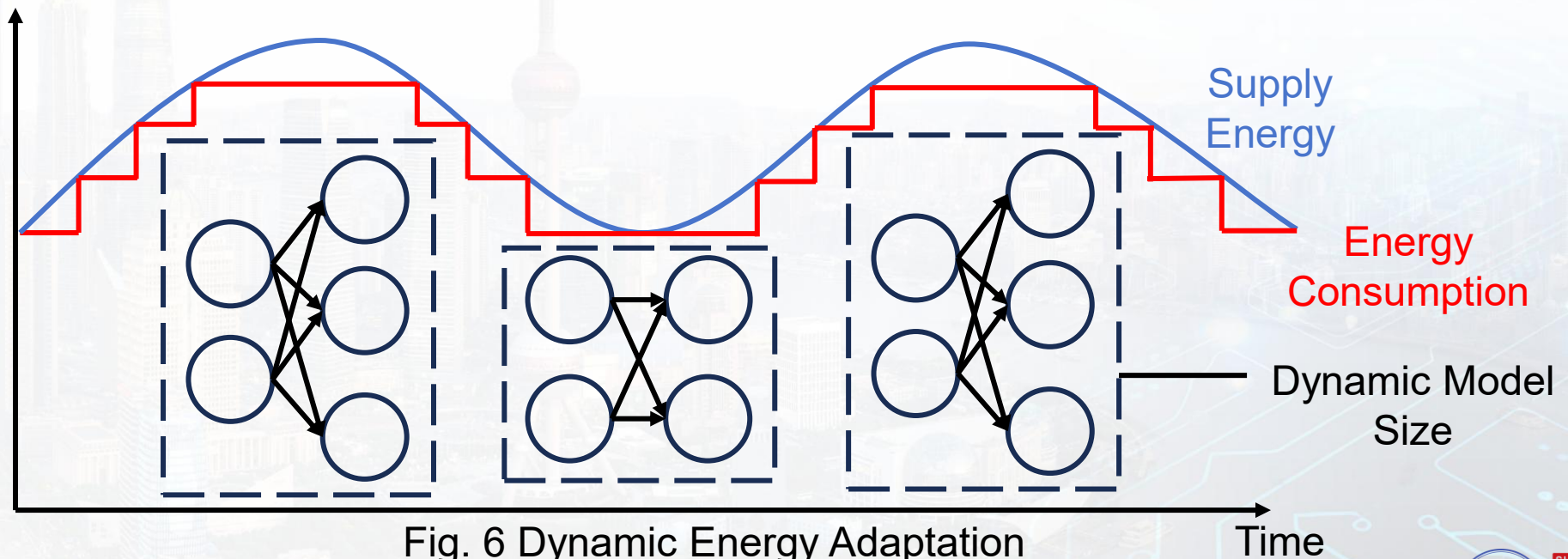
Fig. 5 Simple Energy Reduction

Introduction (cnt.)

Our contributions:

Dynamic energy adaptation framework for Spiking Neural Networks.

Similarity-based neuron pruning to minimize performance loss.



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Overview

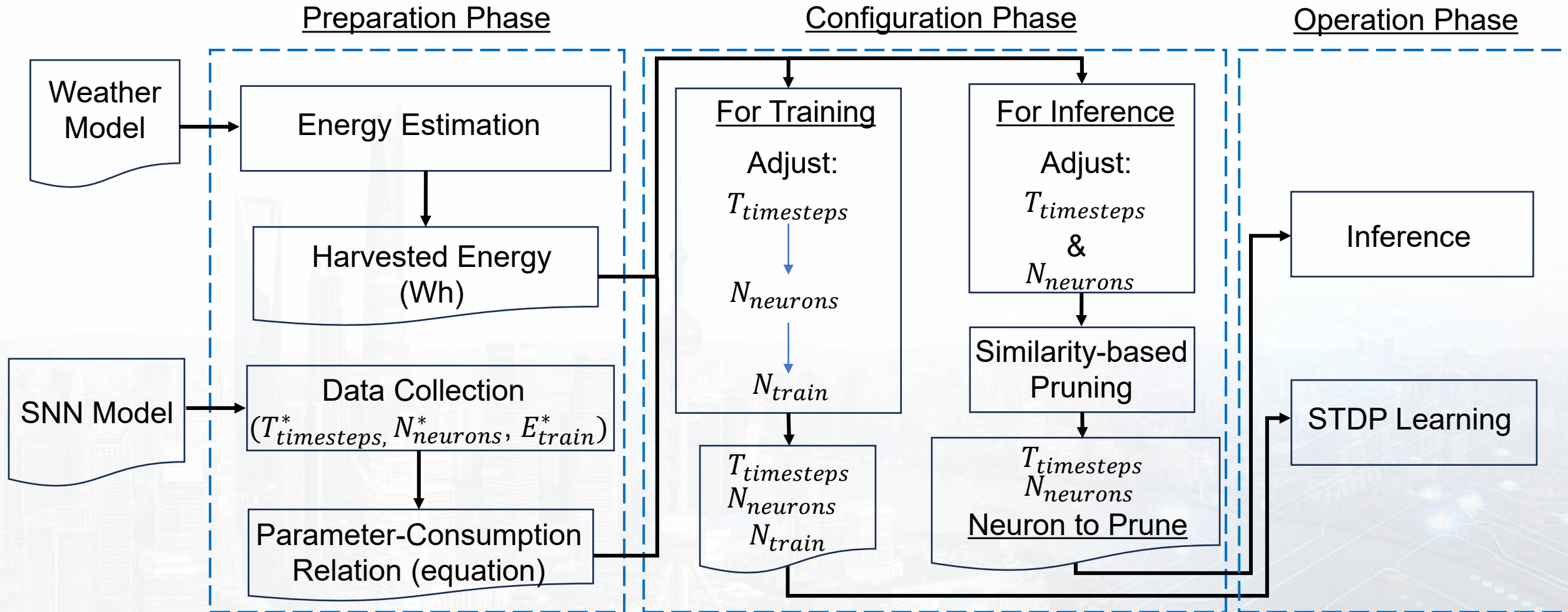


Fig. 7 Framework Overview

Overview

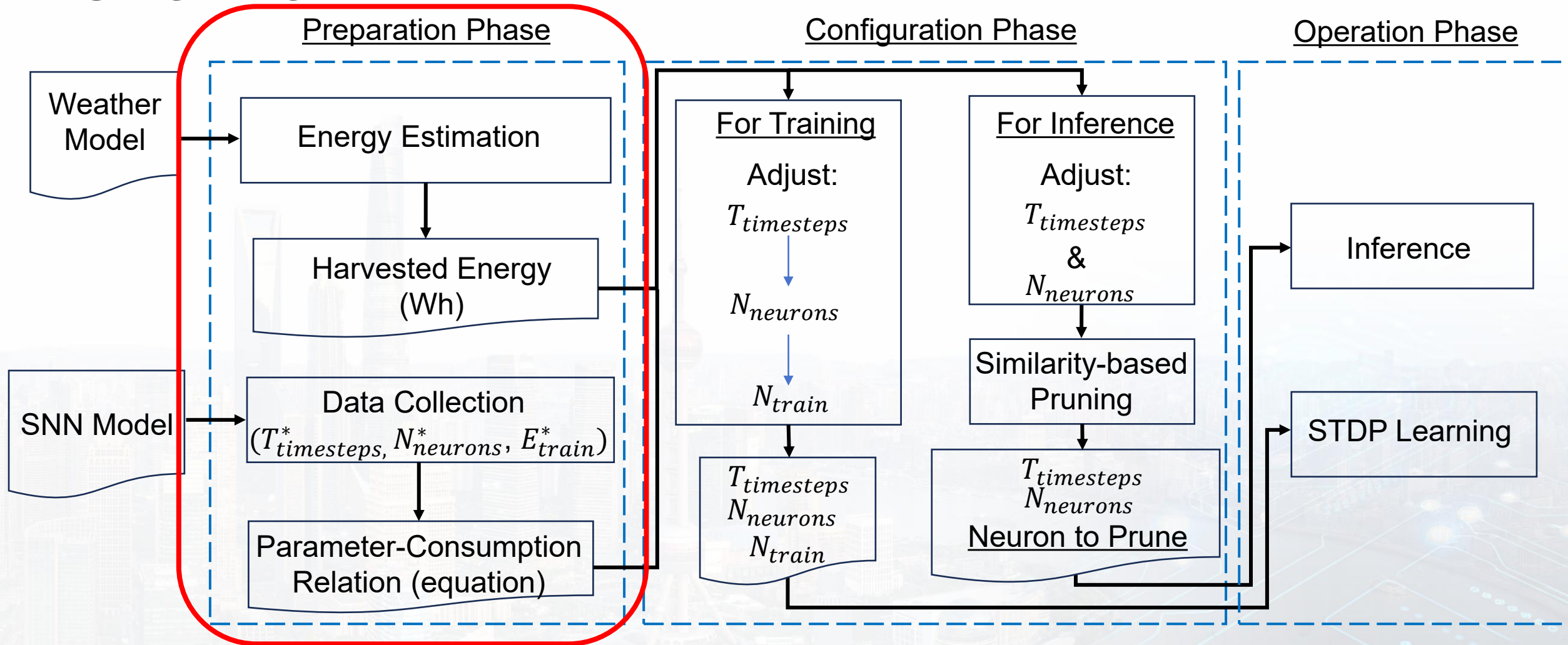


Fig. 7 Framework Overview

Preparation Phase (Energy Estimation)

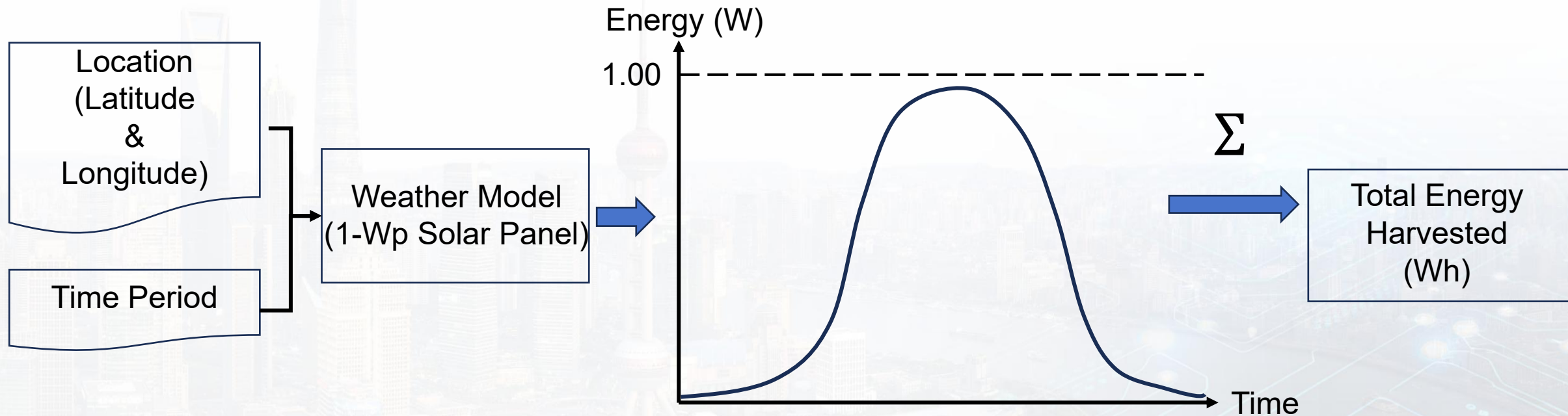


Fig. 8 Estimation of Harvested Energy

Preparation Phase (For Training)

Collect data points

$(T_{timesteps}^*, N_{neurons}(\text{fixed}), E_{train}^*)$



Derive general equation

$$T_{timesteps} = a \times E_{train} + b$$

Linear relation [9]

$(T_{timesteps}(\text{fixed}), N_{neurons}^*, E_{train}^*)$



$$N_{neurons} = a \times E_{train}^d + b \times E_{train} + c$$

Non-linear relation [9]

Preparation Phase (For Inference)

Collect data points

$$(T_{timesteps}^*, N_{neurons}^*, E_{train}^*)$$



Derive general equation

$$E(N_{neurons}, T_{timesteps}) = aN_{neurons}^d + bT_{timesteps} + C$$

Overview

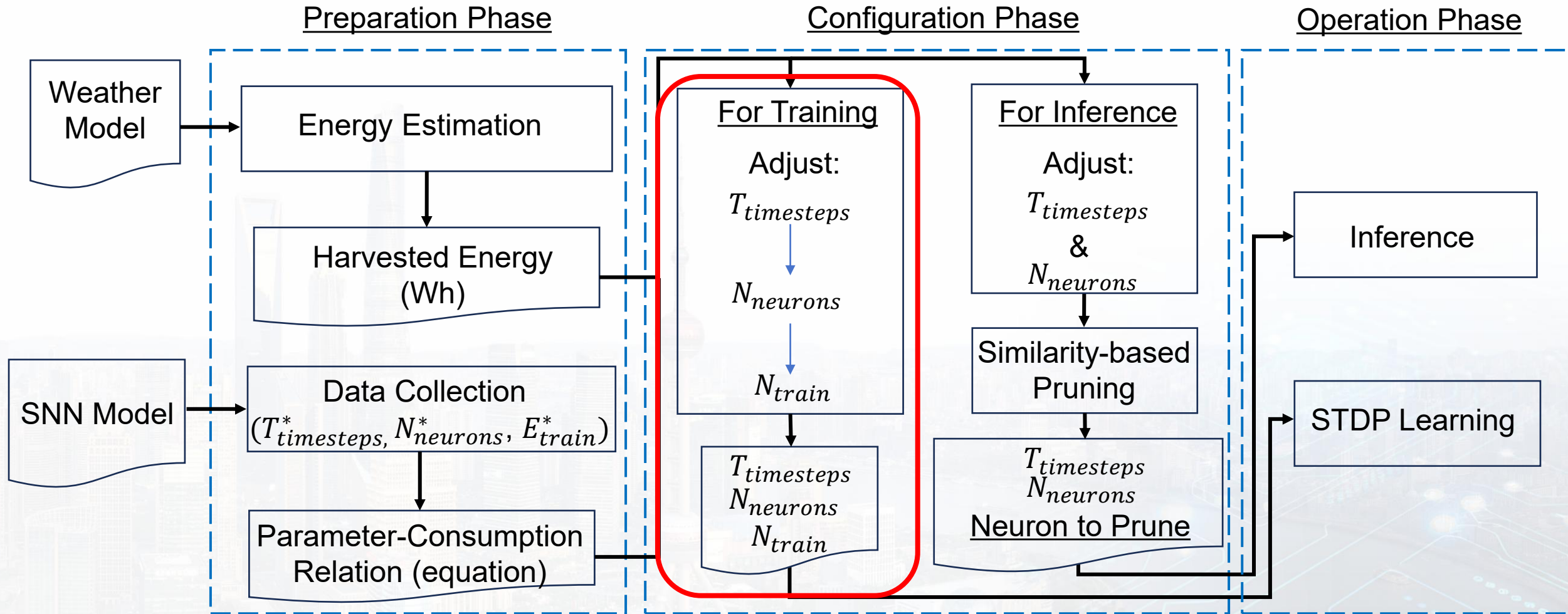


Fig. 7 Framework Overview

Configuration Phase

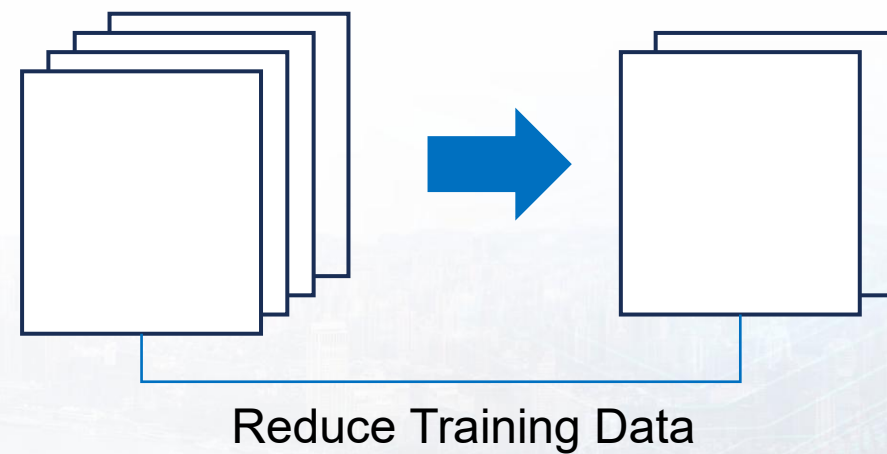
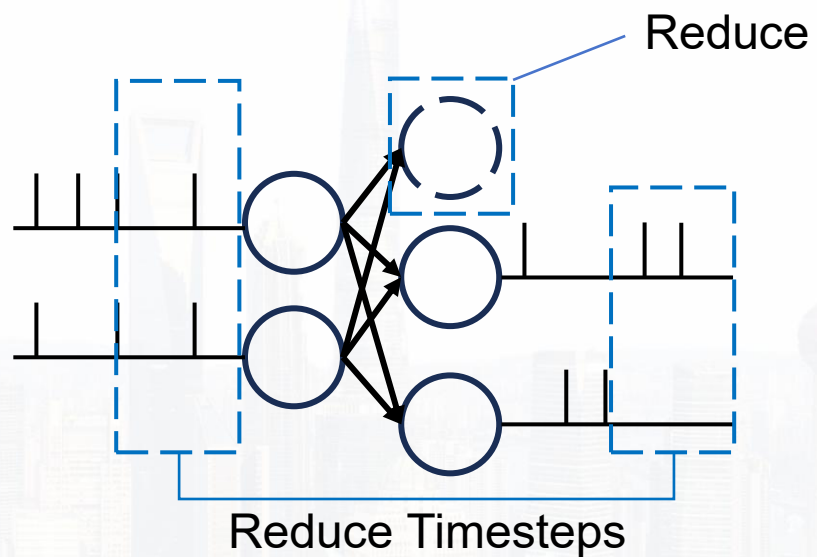


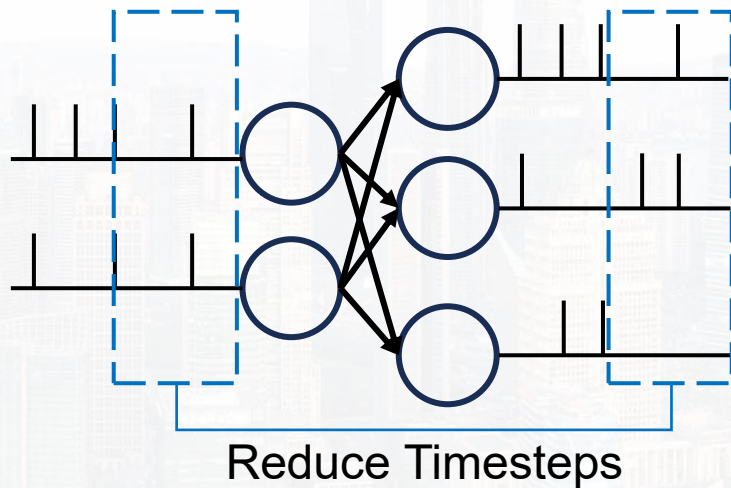
Fig. 12 Timestep Reduction and Pruning

Configuration Phase (Training)

Adjust number of
timesteps

$$T_{timesteps} = a \times \boxed{E_{train}} + b$$

Total Energy
Harvested



If
 $T_{timesteps} < T_{minimum}$
 $T_{timesteps} = T_{minimum}$

Adjust number of
neurons

$$N_{neurons} = a \times \boxed{E_{train}^d} + b \times \boxed{E_{train}} + c$$

Total Energy
Harvested

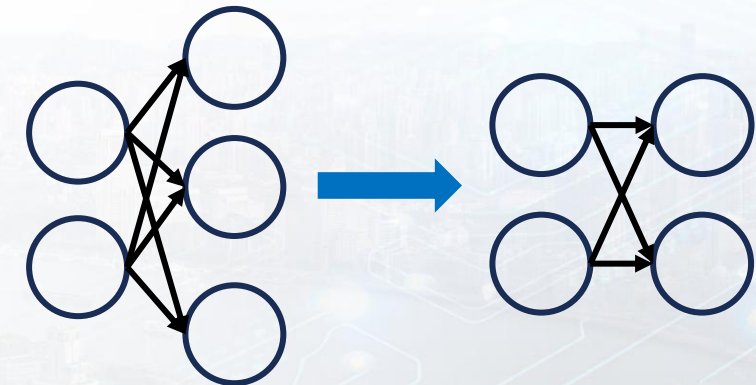
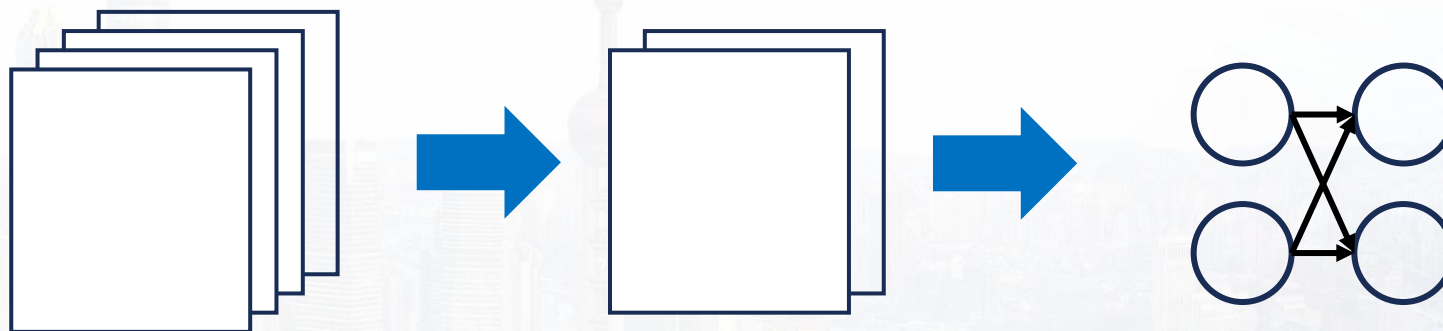


Fig. 9 Timesteps and Neurons Configuration

Configuration Phase (Training)(cnt.)

If
 $N_{neurons} < N_{minimum}$

Reduce Number of
Training Data N_{train}



1. $N_{train} = N_{train} \times \frac{N_{neurons}}{N_{minimum}}$
2. $N_{neurons} = N_{minimum}$

Fig. 10 Training Data Reduction

Overview

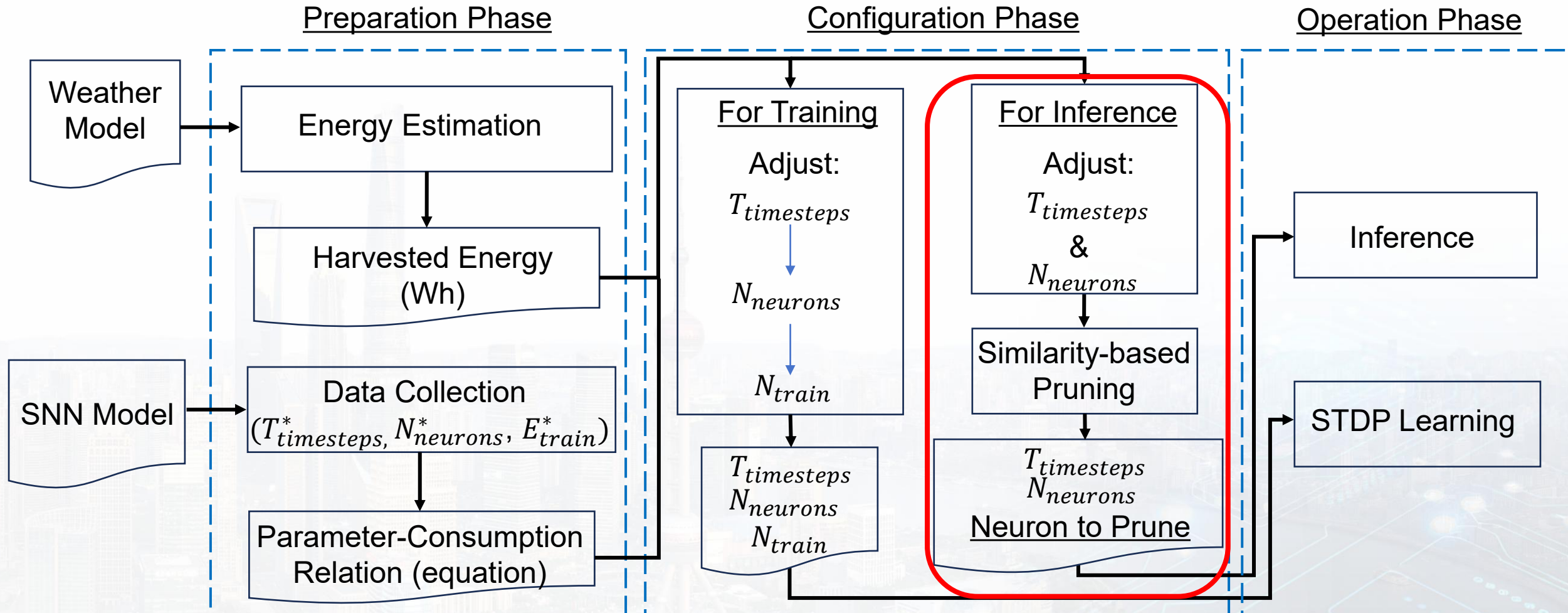


Fig. 7 Framework Overview

Configuration Phase (Inference)

$$E(N_{\text{neurons}}, T_{\text{timesteps}}) = aN_{\text{neurons}}^d + bT_{\text{timesteps}} + C$$



Sequential-Least-Squares-Programming

$$\min_{N_{\text{neurons}}, T_{\text{timesteps}}} \left[\boxed{E_{\text{inference}}} - E(N_{\text{neurons}}, T_{\text{timesteps}}) \right]^2$$

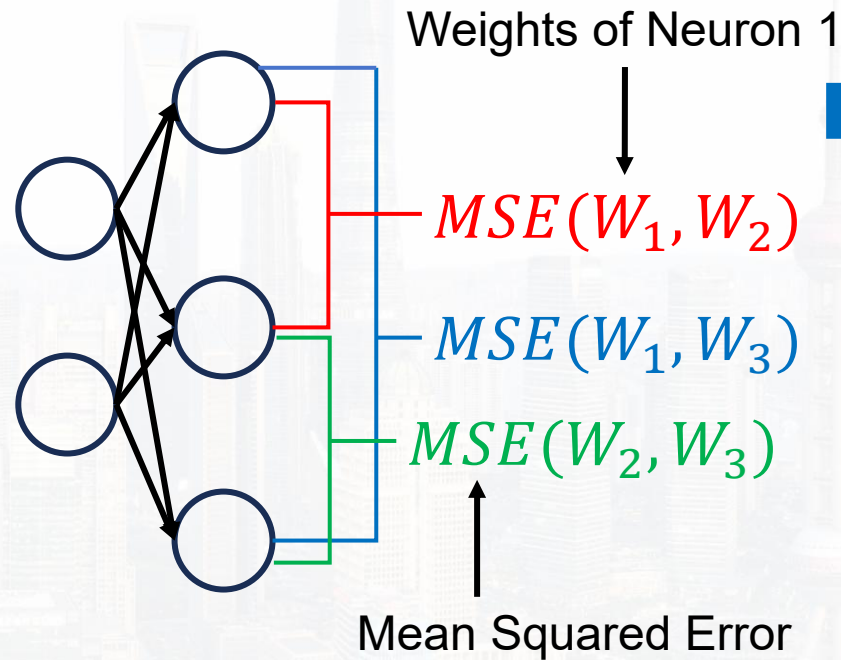
Total Energy Harvested



$T_{\text{timesteps}}, N_{\text{neurons}}$

Configuration Phase (Inference)(cnt.)

1. Measure Distance
(MSE) Between
Neurons



2. Sort in ascending order

$$MSE(W_1, W_2) < MSE(W_1, W_3) < MSE(W_2, W_3)$$

The Most Similar

Neuron To Be Pruned:
Neuron 1 (W_1)

Fig. 11 Similarity-Based Neuron Pruning

Configuration Phase (Inference)(cnt.)

Adjust Timesteps & Prune Neuron

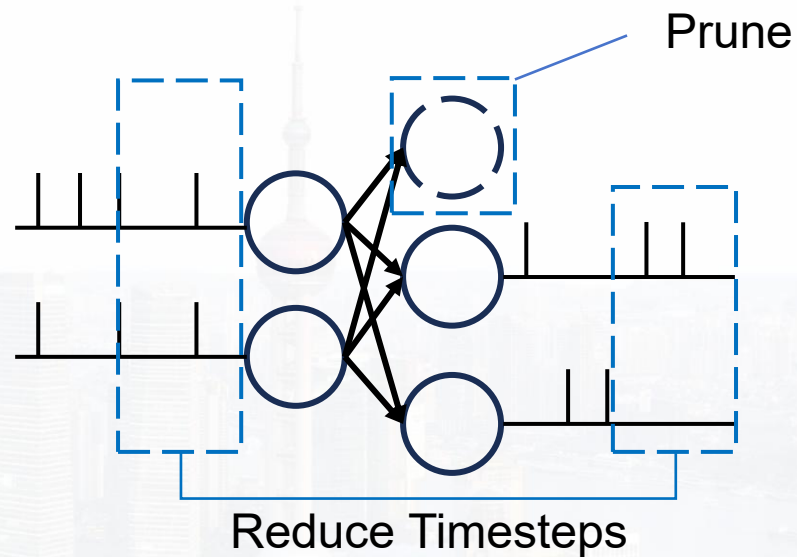


Fig. 12 Timestep Reduction and Pruning

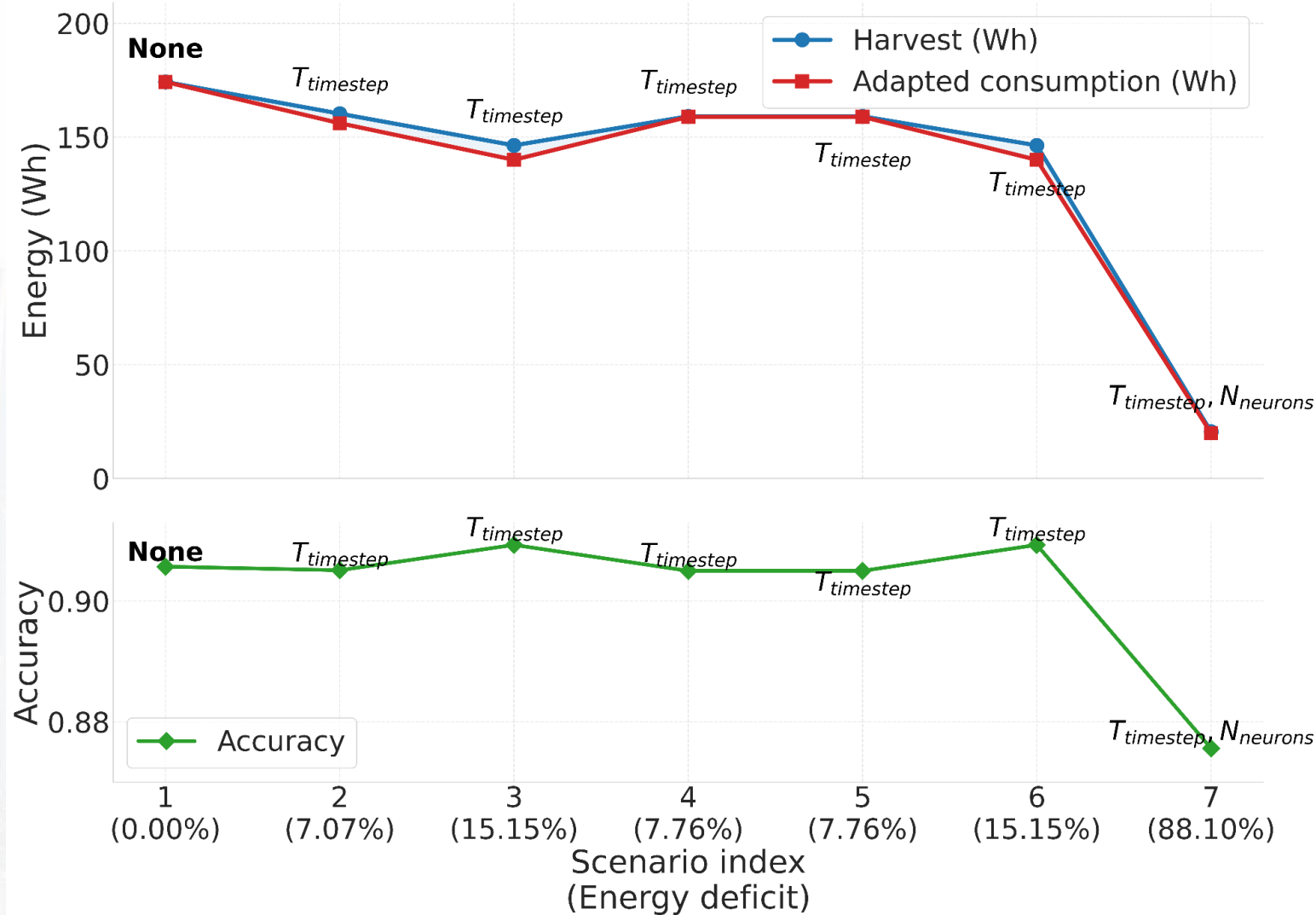
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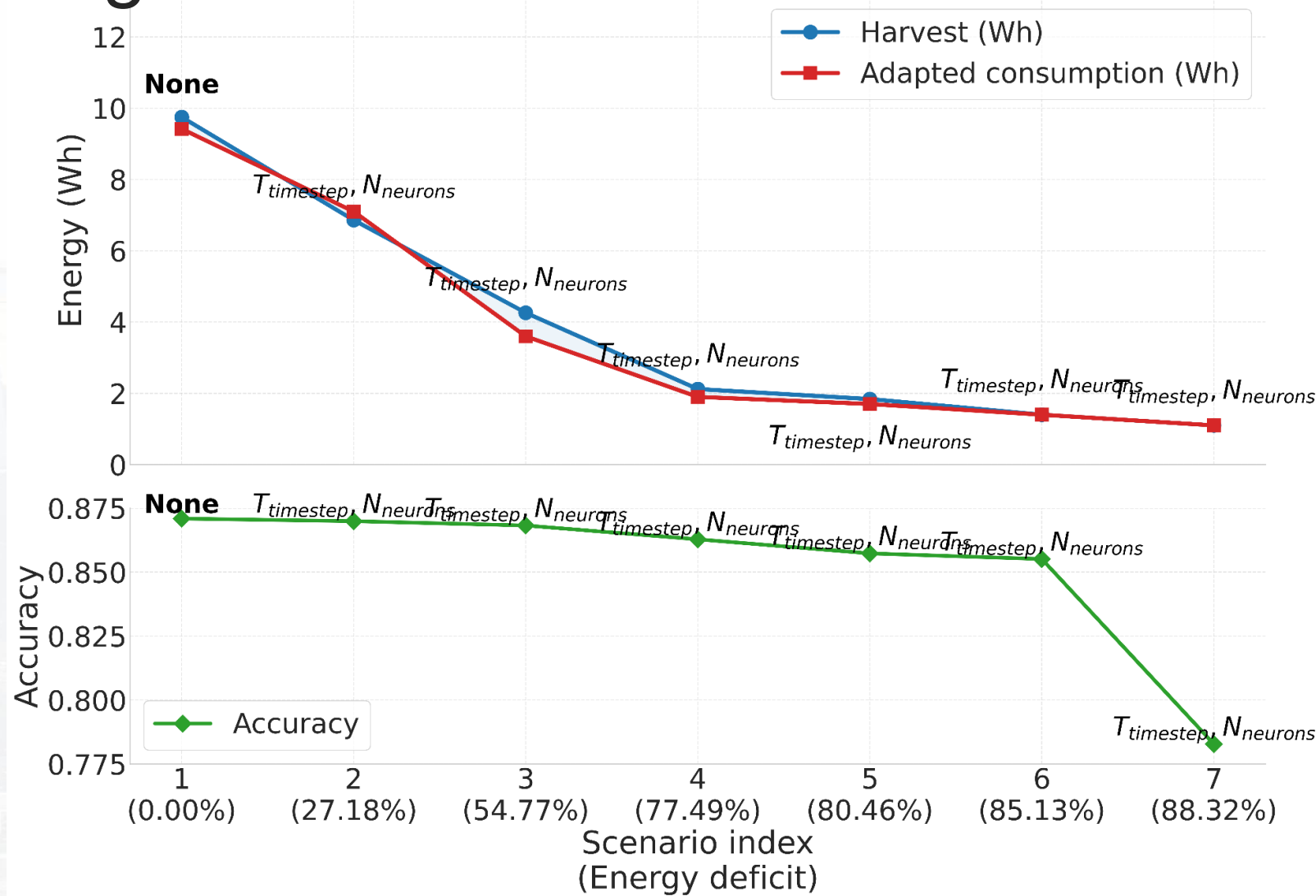
Evaluation Setup

- Implemented in: Python and BindsNet [10] Library
- Device: Raspberry Pi 5
- Model Architecture: Single Layer Feedforward with Lateral Inhibitory Signal
- Training Method: Spike-Timing-Dependent-Plasticity [8]
- Weather Condition: Open-Sourced Quartz-Solar-Forecast [11]
- Battery Simulation: Battery-Dataset-Preprocessing-Code-Library [12]
- Evaluation: Single model & Ensemble Learning [13]

Single 700-neuron Model Training



Single 700-neuron Model Inference



Ensemble Learning

Evaluation Setup: 70-neuron * 10 nodes

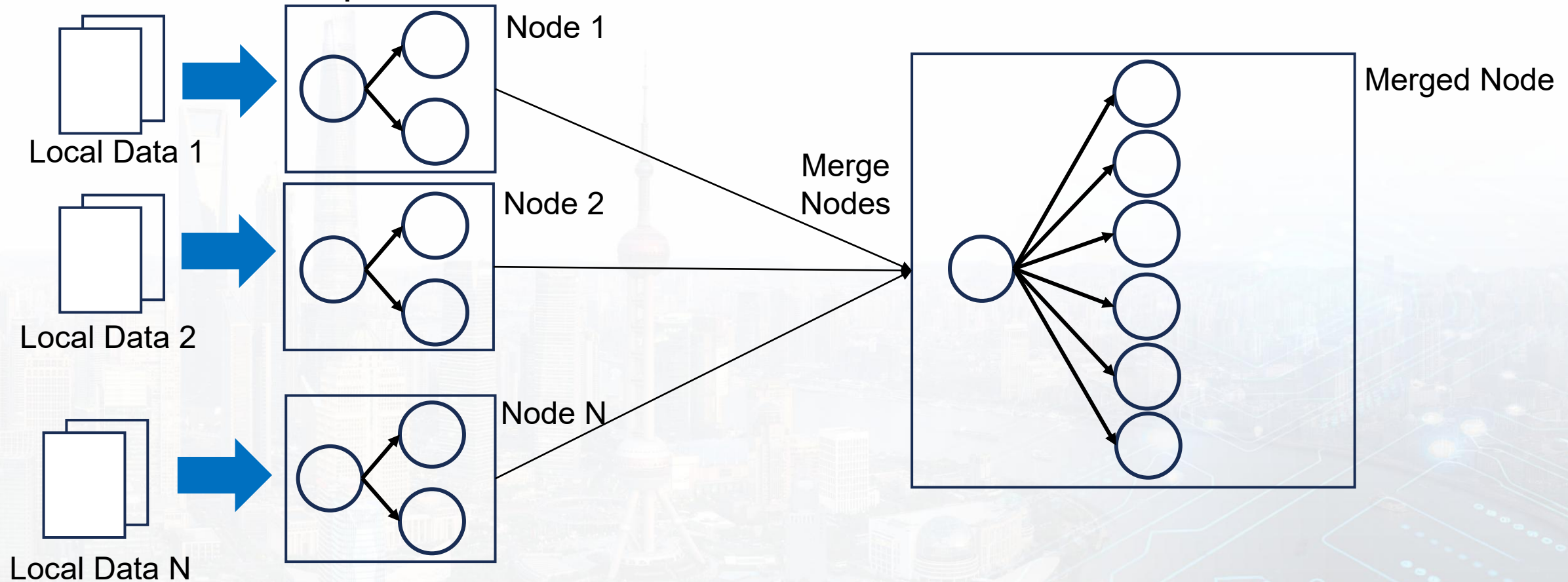
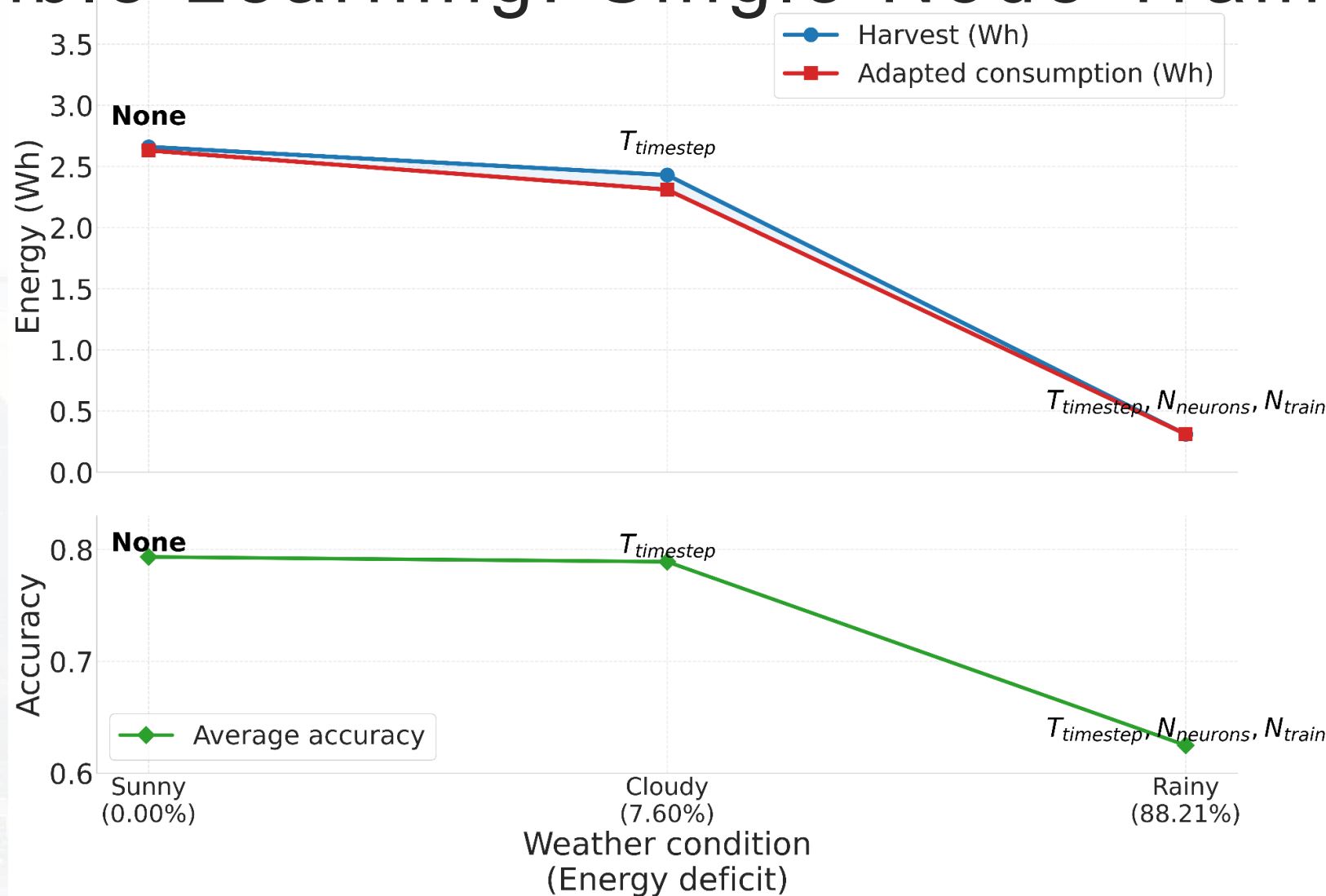
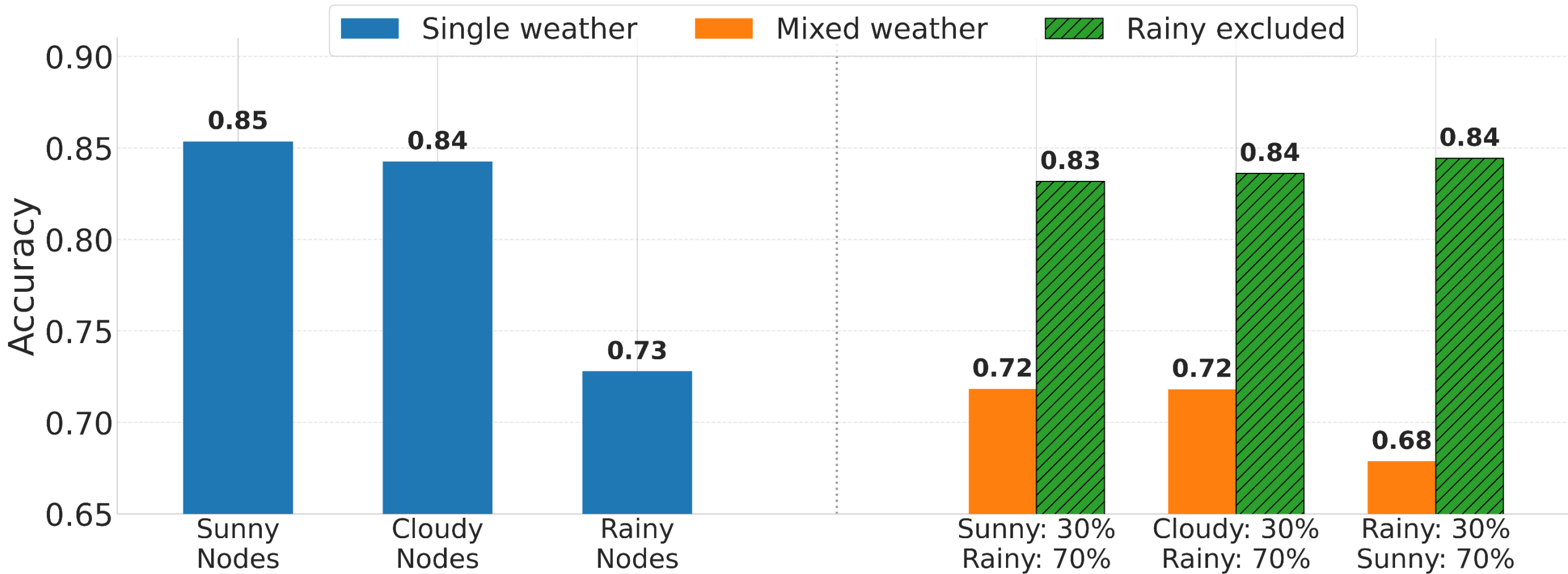


Fig. 13 Ensemble Learning Architecture

Ensemble Learning: Single Node Training



Ensemble Learning: Merged Node



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Conclusion

- GreenMorph framework offers best AI performance under fluctuating energy supply.
- The framework dynamically reduces SNN energy consumption up to 85.13% with 1.5% accuracy loss.
- Future work will expand the parameter adjustment to hardware for further energy reduction.

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Thank you for your attention!